

Environmental Justice in San Francisco

Olivia Knapp, 10/10/26

Hunter's Point Naval Shipyard Superfund Site

In southeast San Francisco lies Hunter's Point Naval Shipyard. Between 1946 to 1969, the naval yard housed the Naval Radiological Defense Laboratory (NRDL). The site leached fuel, pesticides, heavy metals, and radioactive particles into nearby soil, ground water, and surface water. Though the site was added to the National Priorities List in 1989, historical and current inequalities may influence which communities are most affected by the environmental impacts of the site.

Mapping proximity to the site and demographic factors

Load packages and source data

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2

— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(sf)
```

```
Linking to GEOS 3.12.1, GDAL 3.8.4, PROJ 9.4.0; sf_use_s2() is TRUE
```

```
library(tmap)
```

```
# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read('.././../courses/EDS223/ejscreen/
EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb')
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
`/courses/EDS223/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
```

```

    using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS:  WGS 84 / Pseudo-Mercator

```

```

# Filter to a San Francisco county
san_fran <- ejsscreen %>%
  dplyr::filter(CNTY_NAME %in% c("San Francisco County")) |> filter(ACSTOTPOP
> 0)
#Filter out census blocks without residents (bay, Farallon Islands, etc.)

```

Map the Data

```

tm_shape(san_fran) +
  tm_polygons(fill = 'P_PNPL',
              title = "National Percentile Score",
              palette = 'viridis') +
  tm_layout(main.title = "Proximity to Superfund Site",
            fontface = "plain",
            fontfamily = "sans") +
  tm_compass(position = c('right', 'bottom')) +
  tm_scalebar(position = c('right', 'top'))

```

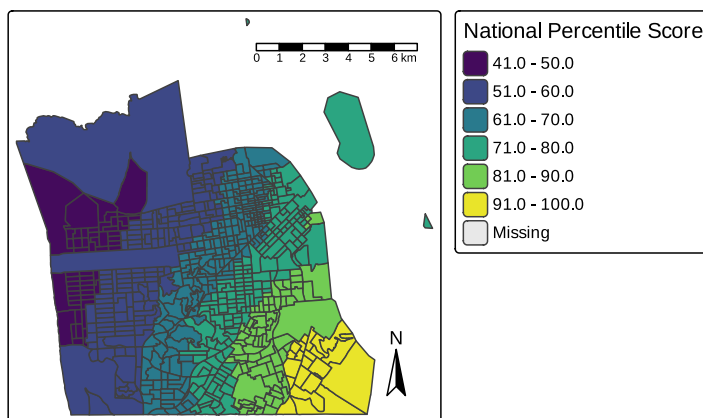
— tmap v3 code detected —

```

[v3->v4] `tm_tm_polygons()`: migrate the argument(s) related to the scale of
the map variable `fill` namely 'palette' (rename to 'values') to fill.scale =
tm_scale(<HERE>).
[v3->v4] `tm_polygons()`: migrate the argument(s) related to the legend of the
map variable `fill` namely 'title' to 'fill.legend = tm_legend(<HERE>)'
[v3->v4] `tm_layout()`: use text.fontfamily instead of fontfamily
[v3->v4] `tm_layout()`: use text.fontface instead of fontface
[v3->v4] `tm_layout()`: use `tm_title()` instead of `tm_layout(main.title = )`

```

Proximity to Superfund Site



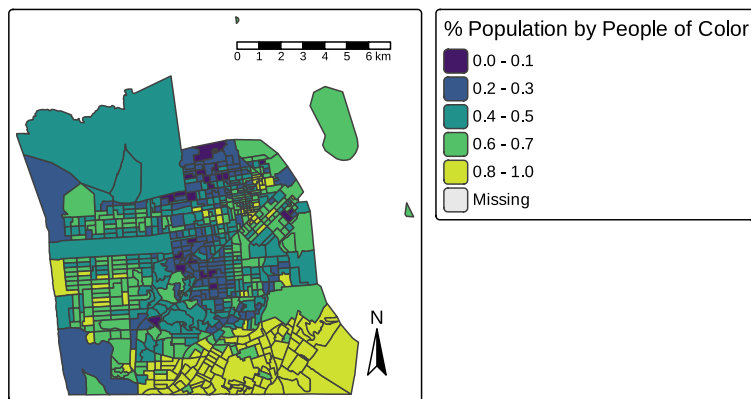
```
tm_shape(san_fran) +
  tm_polygons(fill = 'PEOPCOLORPCT',
              palette = 'viridis',
              title = "% Population by People of Color",
              position = tm_pos_out('right', 'top')) +
  tm_layout(main.title = "People of Color in San Francisco",
            fontface = "plain",
            fontfamily = "sans") +
  tm_compass() +
  tm_scalebar(position = c('right', 'top'))
```

— tmap v3 code detected —

[v3->v4] `tm\_tm\_polygons()`: migrate the argument(s) related to the scale of the map variable `fill` namely 'palette' (rename to 'values') to fill.scale = tm_scale(<HERE>). [v3->v4] `tm\_polygons()`: migrate the argument(s) related to the legend of the map variable `fill` namely 'title' to 'fill.legend = tm_legend(<HERE>)' [tm_polygons()] Argument `position` unknown. [v3->v4] `tm\_layout()`: use text.fontfamily instead of fontfamily [v3->v4] `tm\_layout()`: use text.fontface instead of fontface [v3->v4] `tm\_layout()`: use `tm\_title()` instead of `tm\_layout(main.title = )` [plot mode] fit legend/component: Some legend items or map components do not fit well, and are therefore rescaled.

i Set the tmap option `component.autoscale = FALSE` to disable rescaling.

People of Color in San Francisco



Interpretation

The Hunter's Point Naval Shipyard Superfund Site lies in the southeast corner of San Francisco. Because of the city's relatively small geographic footprint, the entire city scores in at least the 40th national percentile, being closer to a superfund site than 60% of the nation's population. Census blocks close to the site are closer than at least 90% of people living in the United States. Mapping the demographic layout of people of color in San Francisco indicates that the shipyard disproportionately affects people of color. At least 80% of the population closest to the Superfund Site are people of color. To better understand the factors contributing to this inequality, further research can examine the influence of income inequality and housing costs, historical discriminatory practices such as red-lining, and comparisons of health outcomes across the city related to pollution from the site.

References

- City and County of San Francisco. (n.d.). Hunters Point Naval Shipyard - learn more. SF.gov. <https://www.sf.gov/hpns-cleanup-learn>
- United States Environmental Protection Agency (EPA), 2023. EJScreen Technical Documentation.
- United States Environmental Protection Agency (EPA). 2023 version. EJSCREEN. www.epa.gov/ejscreen